Thakur Educational Trust's (Regd.)
THAKUR SHREE DPS COLLEGE OF ENGINEERING & MANAGEMENT
Approved by AICTE & Government of Maharashtra
Affiliated to University of Mumbai

Programme: F.E. (All Branches)

Internal Assessment Test - I (A.Y. 2026-27)

Subject: Applied Physics **NEP- 2020** **Sem: I**

Max. Marks: 15 **Date: 19/10/2025** **Time: 3.00 pm to 4.00 pm**

Note: All questions are compulsory.

Q. No.	Questions	Marks	BL	CO
Q.1	**Attempt any two of the following:**			
1(a)	Apply thin-film interference to explain Anti-reflection coating and state the required condition for minimum thickness with suitable diagram.	3	L2, L3	CO3
1(b)	In a Newton's rings experiment, the diameter of the 18th dark ring is **0.625 cm** and that of the 8th dark ring is **0.412 cm**. The radius of curvature of the plano-convex lens used is **150 cm**. Find the wavelength of light used.	3	L3	CO3
1(c)	Illustrate thin-film interference conditions to determine maxima and minima in reflected and transmitted light, with suitable diagrams.	3	L2	CO3
Q.2	**Attempt any one of the following:**			
2(a)	Derive an expression for numerical aperture of a step-index fiber with diagram, by the principles of total internal reflection.	4	L3	CO2
2(b)	i) Compare step-index and graded-index optical fibers based on their refractive-index profiles. ii) Compute the numerical aperture of a fiber with core refractive index **1.57** and cladding refractive index **1.55**.	4	L3	CO2
Q.3	**Attempt any one of the following:**			
3(a)	Illustrate the construction and working of a He-Ne laser using a labelled energy-level diagram. What is the importance of resonator cavity?	5	L2	CO1
3(b)	i) Explain the terms spontaneous emission, stimulated emission, pumping, and metastable state with suitable diagrams. ii) Define coherence length and time. The coherence length of a light source is **1.5 m**. Determine its coherence time.	5	L2, L3	CO1

BL – Bloom's Taxonomy Levels (L1- Remembering, L2- Understanding, L3– Applying, L4 - Analyzing, L5 - Evaluating, L6 - Creating)
CO – Course Outcomes

Solution Set A.

1(a)	**Apply thin-film interference to explain Anti-reflection coating and state the required condition for minimum thickness with suitable diagram.**	**3 Marks**
	(1) Explanation of antireflection coating. (2) Mathematical Condition for a thin film to act like antireflection coating.	1.5 1.5
1(b)	**In a Newton's rings experiment, the diameter of the 18th dark ring is 0.625 cm and that of the 8th dark ring is 0.412 cm. The radius of curvature of the plano-convex lens used is 150 cm. Find the wavelength of light used.**	**3 Marks**
	Answer: Given Diameter of 18th dark ring, $D_{18} = 0.625$ cm Diameter of 8th dark ring, $D_8 = 0.412$ cm Radius of curvature of lens, $R = 150$ cm For Newton's rings, the diameter of the nth dark ring is given by: $D_n^2 = 4nR\lambda$ For two dark rings: $D_m^2 - D_n^2 = 4R\lambda(m-n)$ Therefore, $\lambda = \frac{D_{18}^2 - D_8^2}{4R(18-8)}$ Substituting the values: $\lambda = \frac{(0.625)^2 - (0.412)^2}{4(150)(10)}$ $\lambda = 3.68135 \times 10^{-5}$ cm Answer: $\boxed{\lambda \approx 368 \text{ nm}}$	1.5 1.5
1(c)	**Illustrate thin-film interference conditions to determine maxima and minima in reflected and transmitted light, with suitable diagrams.**	**3 Marks**
	(1) Diagram for interference in thin film. (2) Conditions for maxima & minima for reflected & transmitted waves.	1 2
2 (a)	**Derive an expression for numerical aperture of a step-index fiber with diagram, by the principles of total internal reflection.**	**4 Marks**
	1. Diagram of wave propagation **2.** Derivation of NA	1 3

2(b)	**i) Compare step-index and graded-index optical fibers based on their refractive-index profiles.**	**2 Marks**
	Any five points of differentiation.	2
	ii) Compute the numerical aperture of a fiber with core refractive index 1.57 and cladding refractive index 1.55.	**2 Marks**
	Given: Core refractive index, $n_1 = 1.57$ Cladding refractive index, $n_2 = 1.55$ Formula: Numerical Aperture (NA) is given by: $NA = \sqrt{(n_1^2 - n_2^2)}$ Calculation: $n_1^2 = (1.57)^2 = 2.4649$ $n_2^2 = (1.55)^2 = 2.4025$ $n_1^2 - n_2^2 = 2.4964 - 2.2201 = 0.0624$ $NA = \sqrt{0.2763} = 0.2498$ Final Answer: Numerical Aperture (NA) ≈ 0.2498	1 1
3 (a)	**Illustrate the construction and working of a He-Ne laser using a labelled energy-level diagram. What is the importance of resonator cavity?**	**5 Marks**
	1. Energy level diagram 2. Construction, working & diagram **3.** Importance of resonator cavity	1 3 1
3 (b)	**i) Explain the terms spontaneous emission, stimulated emission, pumping, and metastable state with suitable diagrams.**	**2.5 Marks**
	1. Definitions & diagrams	2.5
	ii) Define coherence length and time. The coherence length of a light source is 1.5 m. Determine its coherence time.	**2.5 Marks**
	1. Definitions 2. Given: Coherence length, L = 1.5 m The relationship between coherence length (L) and coherence time (τ) is: $L = c.\tau$ where $c = 3 \times 10^8$ m/s is the speed of light. Therefore, $\tau = L/c$ $\tau = 1.5 / (3 \times 10^8)$ $\tau = 0.5 \times 10^{-8}$ s = **$\tau = 5$ ns**	0.5 each